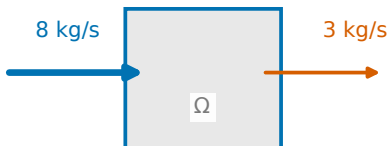
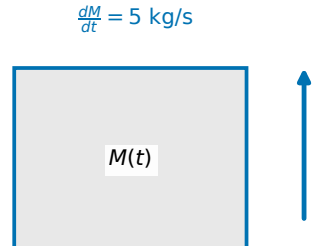
A



$\Phi_{\text{out}} = 3 - 8 = -5$ kg/s

B



$dM/dt = -\Phi_{\text{out}}$